Rutwik Kumar M

linkedin.com/in/rutwikkumarm | Phone: +91 7996957937 | rutwikala@gmail.com | https://github.com/synaptic-scribe

PROFESSIONAL SUMMARY

AI/ML undergraduate with practical experience of machine Learning, deep learning, image processing, and LLM-based systems. Experienced in backend integration and cloud tools, good background working with sponsorship, outreach and event organizing.

EDUCATION

Christ (Deemed to be University) Bangalore, Karnataka
B.Tech in Artificial Intelligence and Machine Learning | Aggregate GPA - 3.35/4 GPA August 2022 – May 2026
Deeksha-Center for Learning Bangalore, Karnataka
Pre-University | Aggregate Percentage - 87.6% July 2021
Shree Swaminarayan Gurukul International School Bangalore, Karnataka
10th Grade | Aggregate Percentage – 80% May 2019

PROJECTS

Autoencoder-Based Anomaly Detection and Segmentation

- Designed a system using an autoencoder trained on MVTec AD hazelnut dataset for image anomaly detection.
- Integrated Alibi identifies defects by using reconstruction error heatmaps detected in the 2048-dimensional latent space of the Outlier VAE.
- Achieved 91% anomaly detection accuracy at 64×64 resolution with model size under 100 KB.

Image Pixelation Detection and Correction

- Developed an image dipixelation system by integrating MobileNetV2 and a upgraded ESPCN model.
- Integrated residual networks and pixel attention for enhanced upscaling.
- Achieved PSNR of 32 dB, SSIM of 81% with an 80 KB lightweight model.

Gemini LangChain Chatbot

- Created a retrieval-Based Chatbot using LangChain and Gemini API.
- Implemented FAISS indexing, PostgreSQL, Gemini embeddings in the framework.

INTERNSHIPS

Shephertz Technologies (*Stipend: 8k per month*) – Intern | April 2025–June 2025

Automation Agent Development & System Integration

- Built an agent using n8n to fetch leads from an Apollo API and automate outreach using Lemlist.
- Created a lead generation workflow with PostgreSQL integration.

Full-Stack app Development

- Containerized PostgreSQL using Docker for local testing.
- Built an Inventory Management System with Tkinter frontend, RESTful API backend, and MongoDB storage.

Intel Unnati Industrial Training - Intern | June 2024–August 2024

- Developed a lightweight (80KB) image depixelation system by integrating MobileNetV2, an improved ESPCN model, residual networks, and pixel attention, achieving a PSNR of 32dB and SSIM of 81%.

Everest IMS – Intern | February 2024–May 2024

Server Utilization forecasting

- Designed and implemented a Python (MLForecaast, XGBoost)-based clustering model using 29,000+ hourly performance data points from 10 servers to analyze, determining the optimum number of clusters for workload partition.
- Investigated various methods to find the optimal number of clusters using different types of clustering and evaluation metics, such as K-means clustering and Calinski-Harabasz Index.

Document retrieval chatbot

- Developed a retrieval based generation (RAG) chatbot using LangChain, Gemini API, and FAISS, which has the ability to quickly perform semantic search and context-sensitive responses, processing thousands of document chunks in real time to answer queries.

CERTIFICATIONS & ACHIEVEMENTS

NASA Space Apps Challenge – Qualifiers for Nationals August-2025
Certified Cloud Practitioner – AWS July 2025-July 2028
LangChain- Develop LLM powered applications with LangChain – Udemy May 2024
Data Science for Engineers – NPTEL March 2024

SKILLS

- Languages: Python, SQL
- Machine Learning & Deep Learning: Autoencoders, Variational Autoencoders (VAE), ESPCN, MobileNetV2, XGBoost, MLForecast, K-Means Clustering, RAG, Gemini API, Computer Vision.
- Tools & Frameworks: LangChain, Streamlit, Chainlit, Tkinter, FAISS, RESTful APIs, Docker, N8N.
- Databases: PostgreSQL, MongoDB
- Cloud: AWS (Certified Cloud Practitioner)
- Soft Skills: Technical Documentation, Team Collaboration, Event Coordination, Strategic Planning.

CO-CURRICULAR & EXTRACURRICULAR

- Department representative & Documentation Head – Christ Incubation Centre.
- Event Planning Vice head - CUESTIC
- Core member - Sponsorship & Publicity team, MAGNOVITE-2024
- Secretary - IEEE-MTT Society (2023-2024)
- Secretary - CUESTIC (2023)
- Member - Sponsorship and Publicity team, MAGNOVITE-2023

LANGUAGES

English | Hindi | Telugu | Kannada